

VAUXHALL, AB, Canada

WMO: 712510

Lat: 50.053N

Lon: 112.126W

Elev: 779

StdP: 92.31

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.1	-26.6	-33.5	0.2	-30.0	-29.7	0.3	-26.5	15.6	4.7	14.1	3.1	1.9	210	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	31.9	18.0	29.9	17.4	28.1	16.8	20.1	27.8	18.9	26.7	17.9	25.6	4.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	13.9	23.1	16.2	12.7	22.0	15.1	11.8	20.8	61.1	27.6	57.1	26.6	53.5	25.5	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.5	9.9	DB	-35.0	35.3	4.0	1.7	-37.9	36.6	-40.3	37.6	-42.6	38.6	-45.5	39.8	(4)
(5)			WB	-35.1	22.3	4.0	1.6	-38.0	23.5	-40.4	24.4	-42.6	25.3	-45.5	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-8.1	-6.4	-0.9	5.9	11.5	15.5	19.0	18.0	13.1	6.2	-1.3	-7.1	(6)	
	DBStd	11.53	9.50	8.86	8.16	4.97	4.43	3.15	2.90	3.23	4.39	5.09	7.77	8.69	(7)	
	HDD10.0	2558	562	458	337	138	36	2	0	1	19	136	342	529	(8)	
	HDD18.3	4776	820	692	594	373	215	94	27	45	162	377	589	788	(9)	
	CDD10.0	927	0	0	1	14	82	168	279	250	113	18	2	0	(10)	
	CDD18.3	103	0	0	0	0	3	10	47	37	7	0	0	0	(11)	
	CDH23.3	2067	0	0	1	12	117	205	783	714	228	8	0	0	(12)	
	CDH26.7	707	0	0	0	1	27	46	283	274	74	1	0	0	(13)	
(14) Wind	WSAvg	4.1	4.3	4.0	4.5	4.7	4.3	3.9	3.3	3.2	3.6	4.4	4.4	4.3	(14)	
(15) Precipitation	PrecAvg	351	18	13	20	34	41	60	35	38	39	15	15	18	(15)	
	PrecMax	489	42	28	40	113	133	151	80	98	164	44	44	69	(16)	
	PrecMin	249	1	0	2	0	6	10	2	6	4	0	3	2	(17)	
	PrecStd	85	13	7	12	27	27	37	22	23	35	12	11	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	13.7	18.2	25.0	29.7	30.9	34.1	34.3	32.1	24.4	17.8	11.9	(19)	
		MCWB	6.9	6.4	8.7	11.7	14.9	19.1	18.9	17.7	15.6	12.9	9.9	6.0	(20)	
	2%	DB	8.6	9.9	15.2	21.3	26.7	27.9	31.9	31.9	28.8	20.7	14.0	8.8	(21)	
		MCWB	4.5	4.3	7.7	10.5	13.9	16.7	19.3	17.2	15.1	11.5	7.8	4.2	(22)	
	5%	DB	6.6	7.2	12.7	18.2	23.7	25.7	29.8	29.9	26.0	17.8	11.5	6.6	(23)	
		MCWB	3.3	2.9	6.2	8.8	12.8	15.8	18.6	17.1	14.3	10.1	6.3	2.9	(24)	
	10%	DB	4.5	4.9	9.9	15.3	21.1	23.7	27.8	27.6	23.0	15.1	9.0	4.6	(25)	
		MCWB	2.0	1.7	4.7	7.5	11.6	14.9	18.0	16.7	13.3	8.5	4.5	1.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	7.1	9.9	12.8	16.5	20.3	22.8	20.5	17.5	14.0	10.6	6.8	(27)	
		MCDB	10.7	13.2	16.7	22.4	26.3	27.8	29.5	27.9	27.8	22.4	16.6	11.0	(28)	
	2%	WB	4.9	4.7	8.1	11.0	14.8	18.3	20.7	19.2	15.9	11.9	8.2	4.6	(29)	
		MCDB	8.3	9.0	14.2	19.8	23.5	25.3	28.2	27.4	26.1	19.7	13.4	8.1	(30)	
	5%	WB	3.3	3.2	6.5	9.5	13.6	17.0	19.6	18.1	14.8	10.3	6.5	3.2	(31)	
		MCDB	6.2	6.8	11.9	17.1	22.2	23.4	27.3	26.7	24.1	16.9	11.2	6.3	(32)	
	10%	WB	2.0	1.8	4.9	8.0	12.3	15.7	18.6	17.2	13.6	9.0	4.8	1.7	(33)	
		MCDB	4.4	4.7	9.6	14.4	20.0	21.9	26.0	25.7	21.9	14.5	8.5	4.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	12.0	12.3	13.4	13.7	12.8	15.3	15.9	15.1	13.4	11.6	11.1	(35)	
		MCDBR	11.6	14.0	16.1	19.4	18.7	17.1	19.2	20.8	21.5	18.0	14.3	11.7	(36)	
	5% WB	MCWBR	8.4	9.3	9.5	9.8	8.2	7.9	8.4	8.3	9.4	9.7	8.8	8.2	(37)	
		MCWBR	11.5	13.5	15.5	17.6	16.4	15.3	17.3	17.6	19.4	17.1	13.8	11.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.264	0.272	0.296	0.338	0.354	0.349	0.350	0.370	0.327	0.293	0.266	0.251	(40)	
		taud	2.438	2.472	2.455	2.343	2.345	2.410	2.421	2.343	2.441	2.512	2.489	2.431	(41)	
	Ebn at Noon		801	889	920	907	902	908	901	863	871	849	795	765	(42)	
		Edh at Noon	59	74	90	113	118	112	109	112	91	71	56	51	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.23	2.22	3.53	4.88	5.80	6.37	6.95	5.56	3.96	2.45	1.36	0.95	(44)	
	RadStd		0.11	0.19	0.27	0.34	0.43	0.41	0.35	0.35	0.34	0.26	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (74 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page